

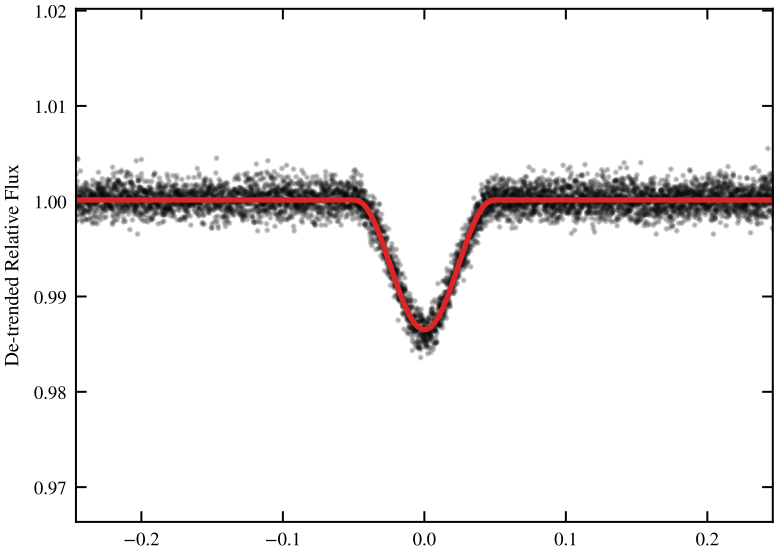
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 469782185 | RA: 292.39150° | Dec: -37.30401° | Sectors: 67

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	1.2093 ± 1.83e-05
Rp/R★	0.35888 ± 0.0742
Rp (R⊕)	48.737 ± 10.4
b	1.1181 ± 0.0848
T14 (h)	2.364 ± 0.034
a (AU)	0.018839 ± 0.00114
Teq (K)	2142.5 ± 47.8

Primary Phased Transit



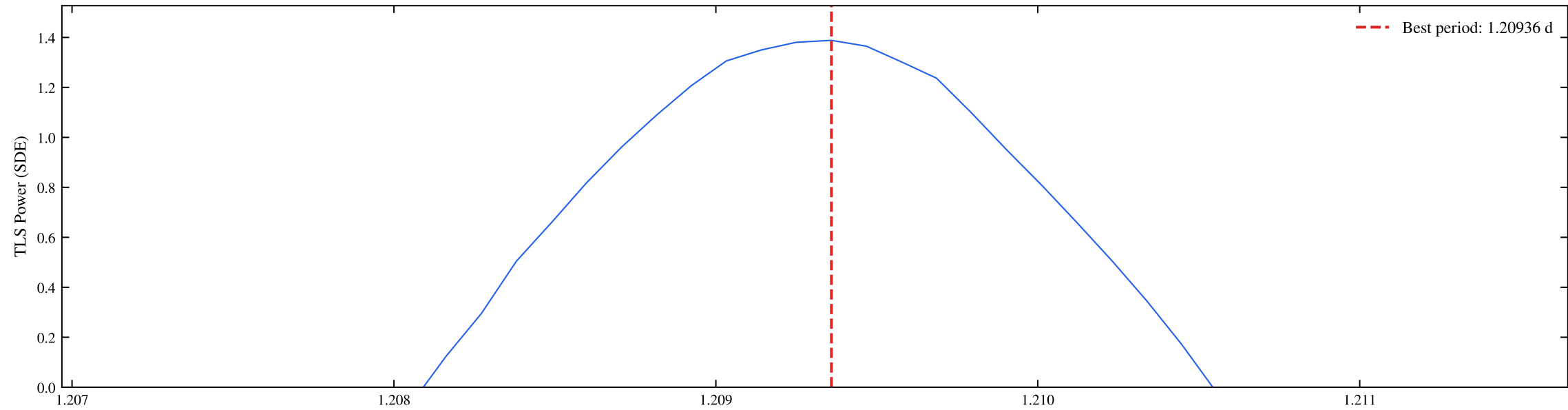
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	1.20936	1.209315 ± 0.00002
P1 Epoch (BTJD)	N/A	3127.3471 ± 0.0002
Max R-hat (R)	N/A	1.0021
Min ESS	N/A	5240
MCMC Divergences	N/A	13

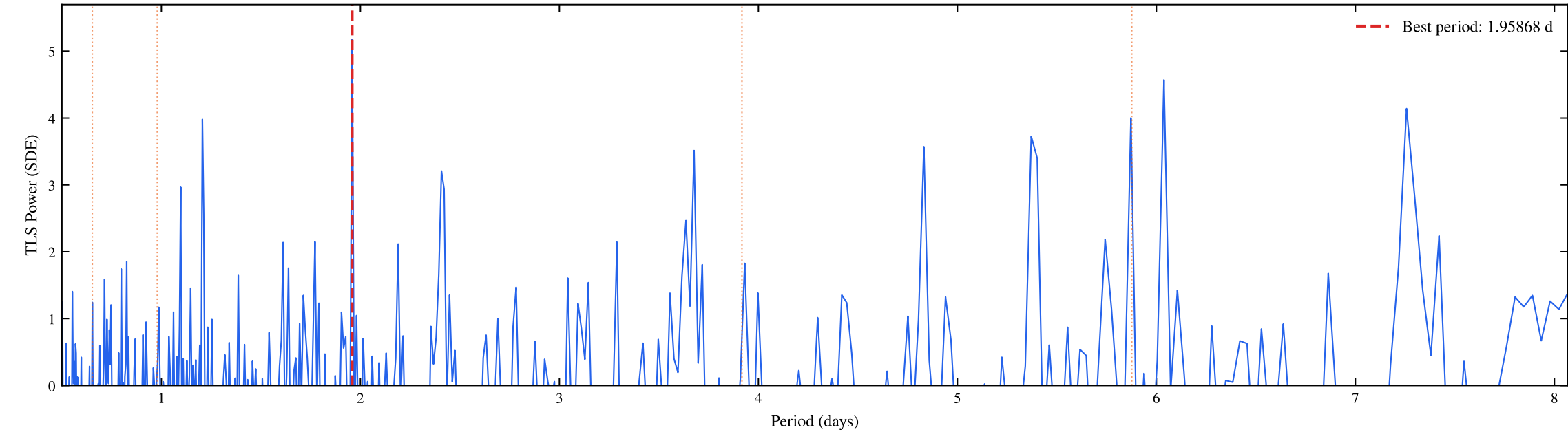
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.242 ± 0.071	VizieR TIC8
M★ (M☉)	0.96 ± 0.11	VizieR TIC8
Teff (K)	5471 ± 109	VizieR TIC8
log g (dex)	4.232 ± 0.0964	VizieR TIC8
[Fe/H]	-1.069 ± 0.0113	Gaia DR3
ρ★ (g/cm³)	0.7064 ± 0.146	Derived

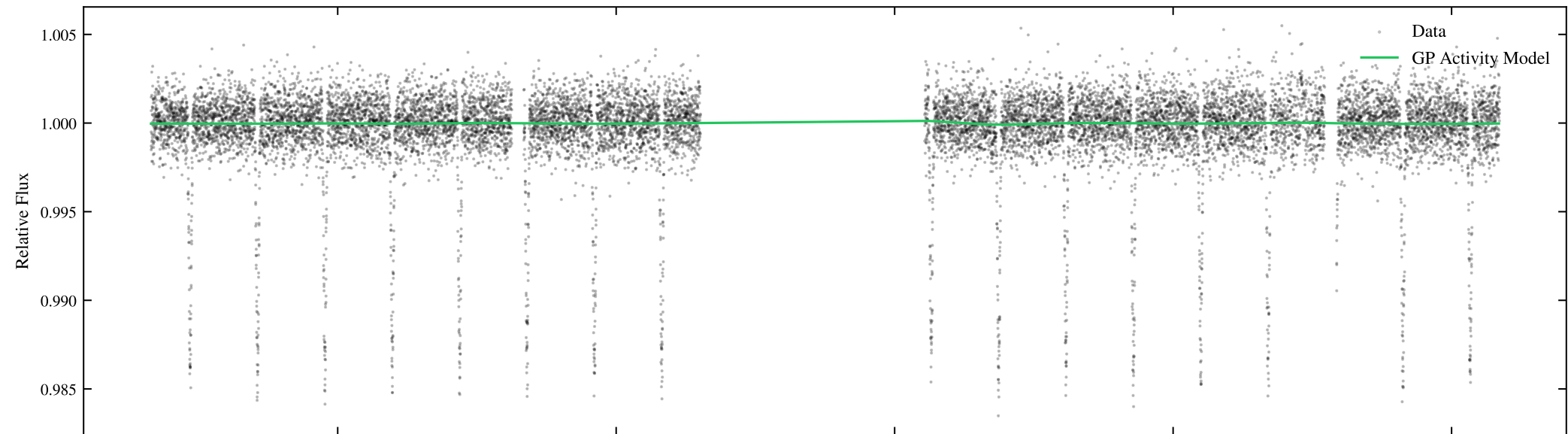
TIC 469782185 | Sectors: 67 | TLS Periodogram (Planet 1 Search) (SDE = 1.39)



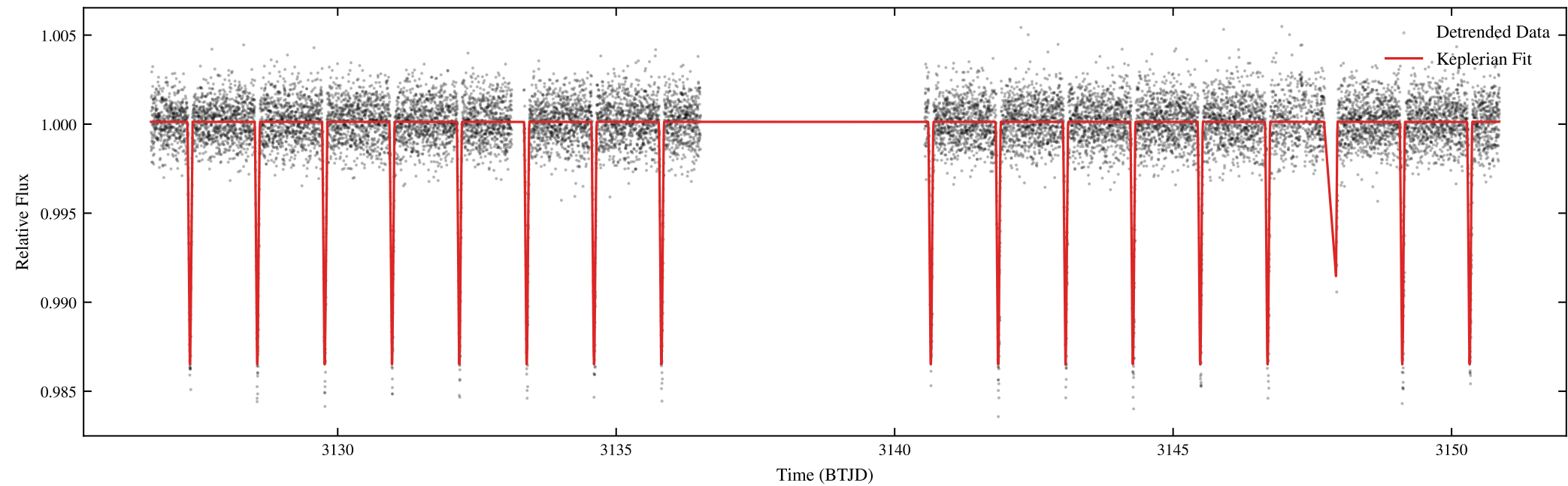
TIC 469782185 | Sectors: 67 | TLS Periodogram (Planet 2 Search) (SDE = 5.18)



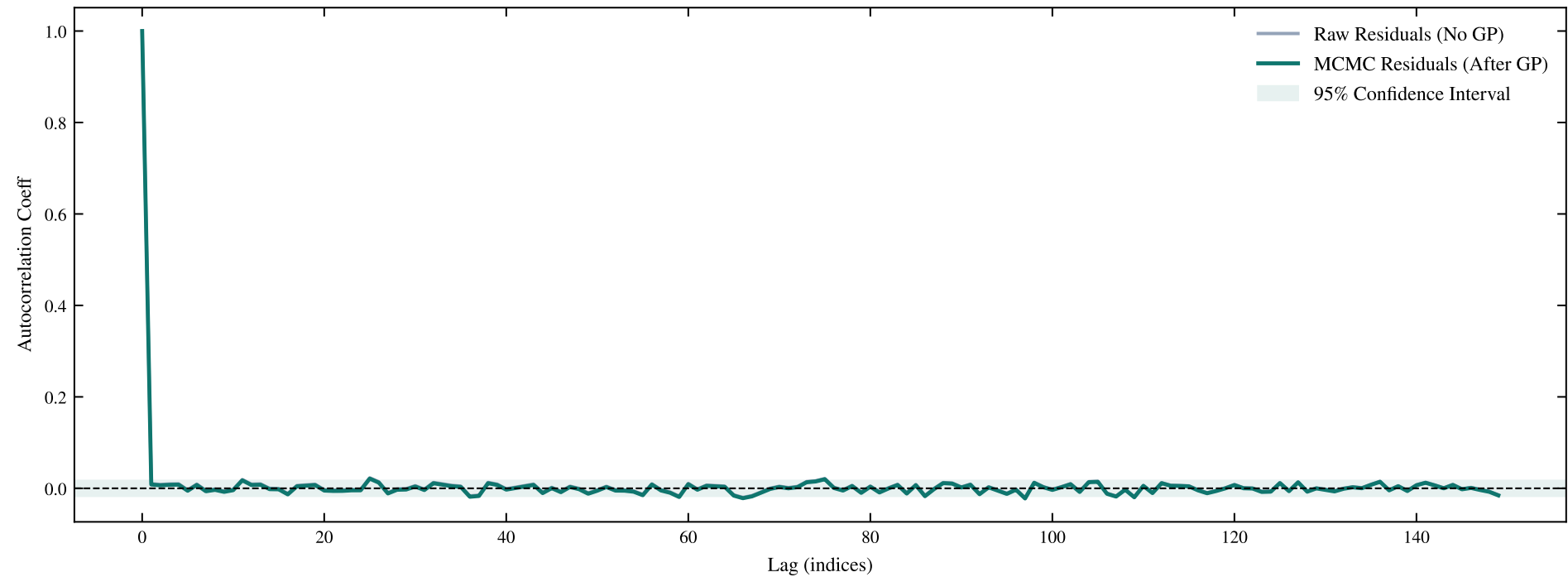
TIC 469782185 | Sectors: 67 | Bayesian GP Detrending (Time Domain)



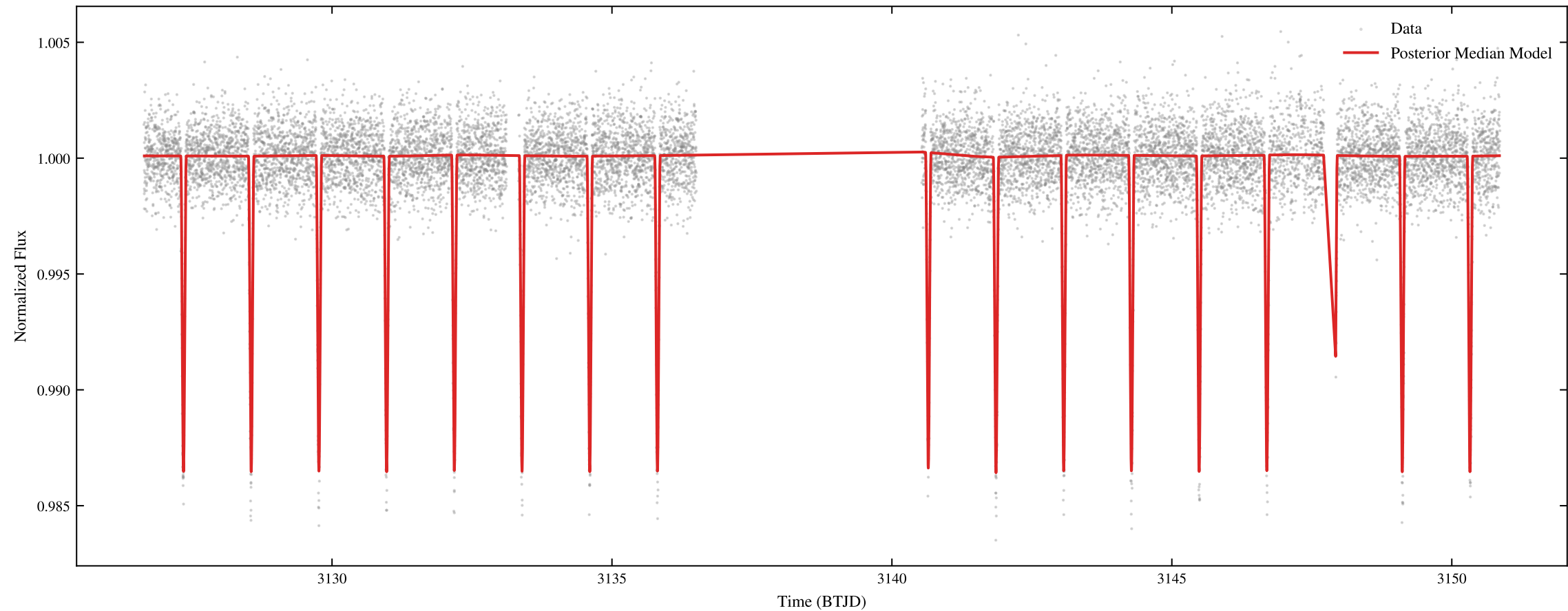
Detrended Light Curve & Best-Fit Transit Model

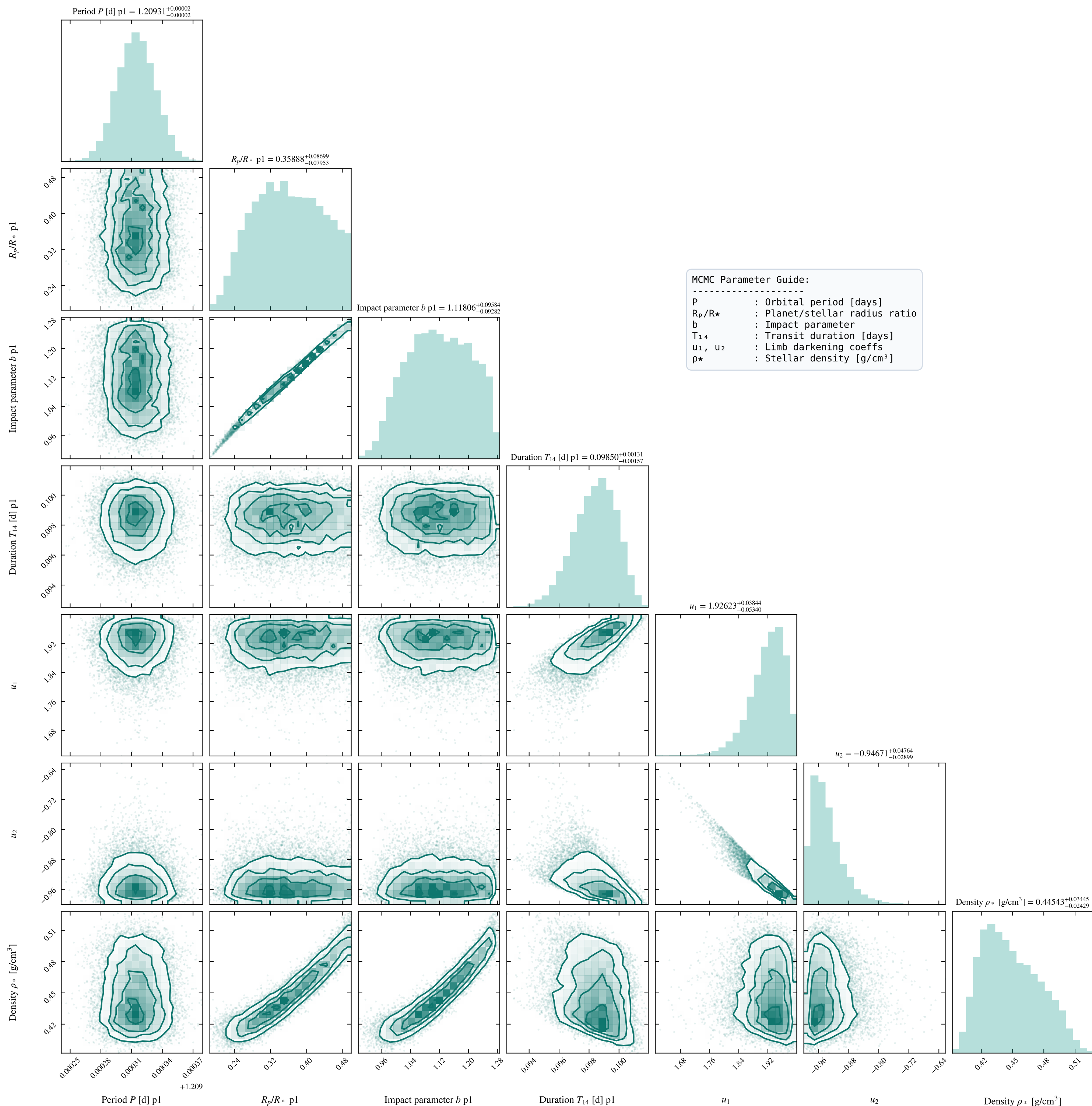


TIC 469782185 | Sectors: 67 | GP Residuals Autocorrelation Function (ACF)

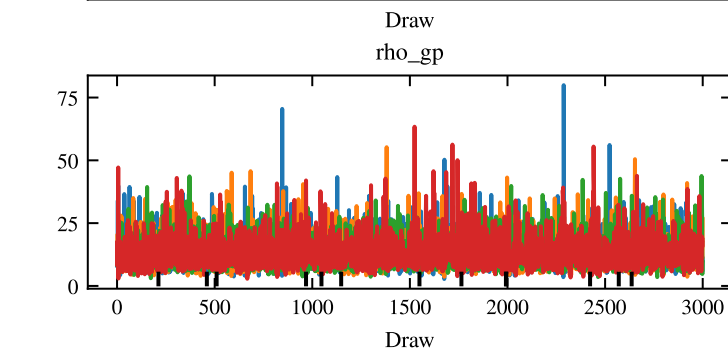
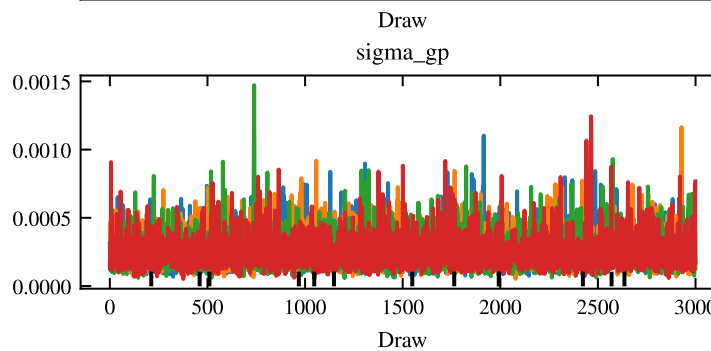
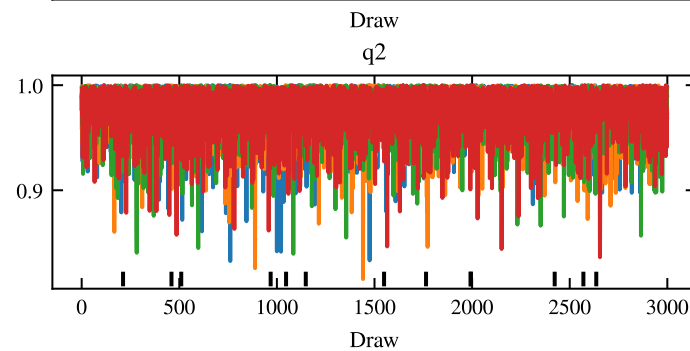
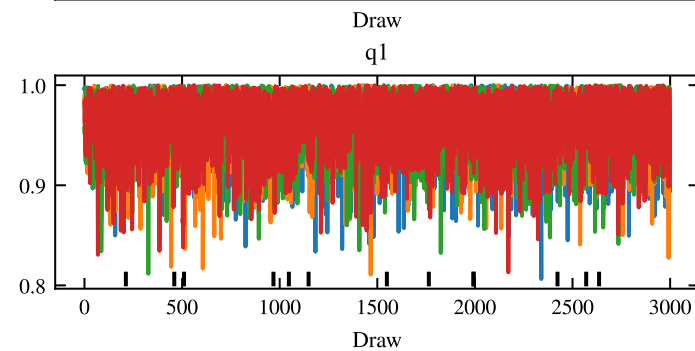
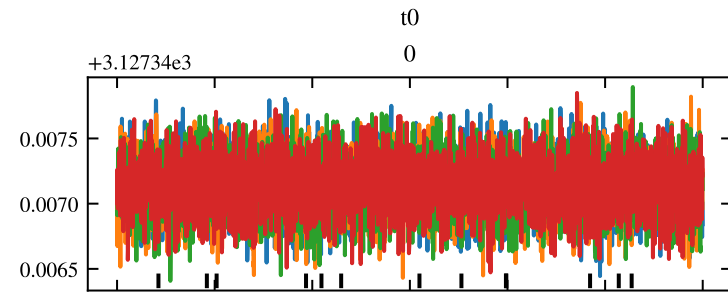
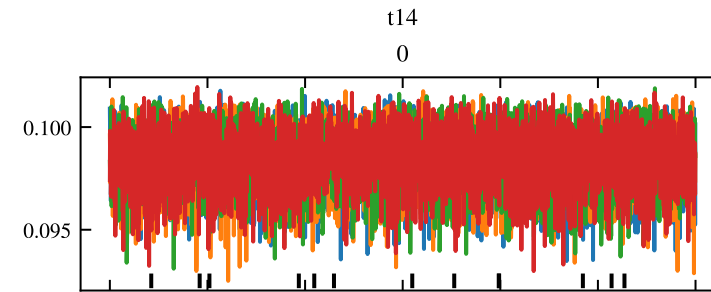
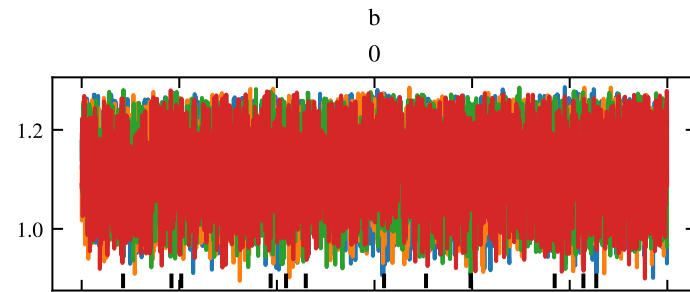
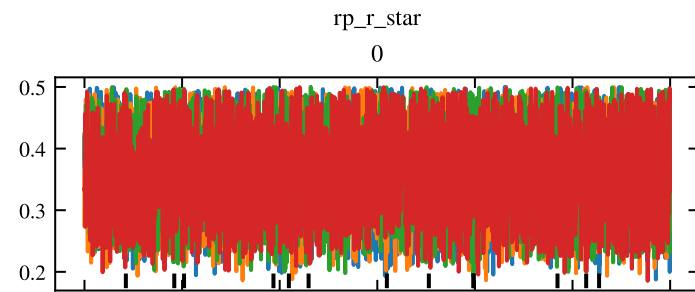


TIC 469782185 | Sectors: 67 | MCMC Posterior Predictive Check

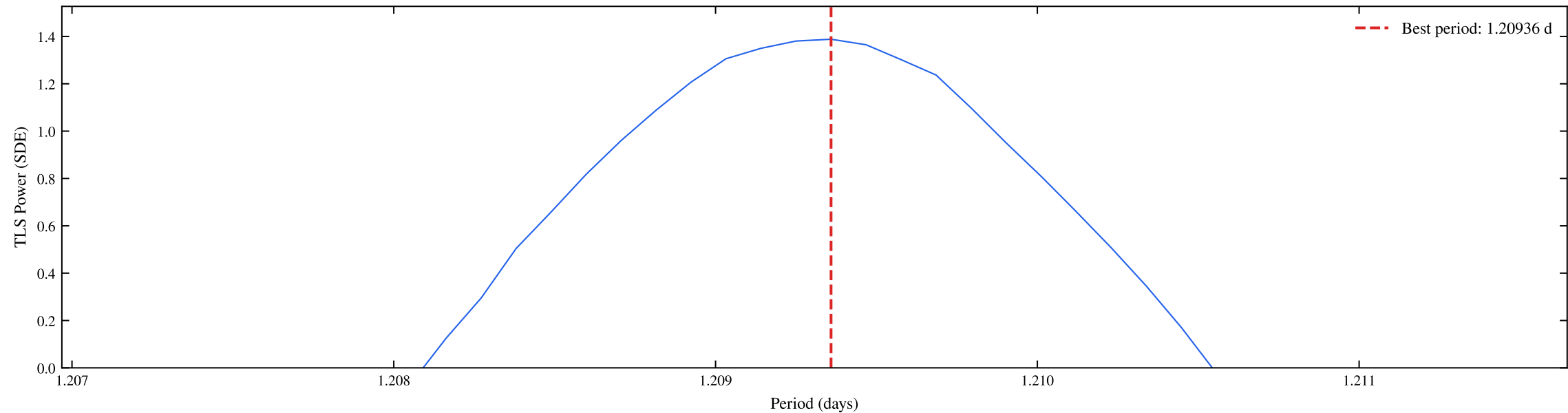




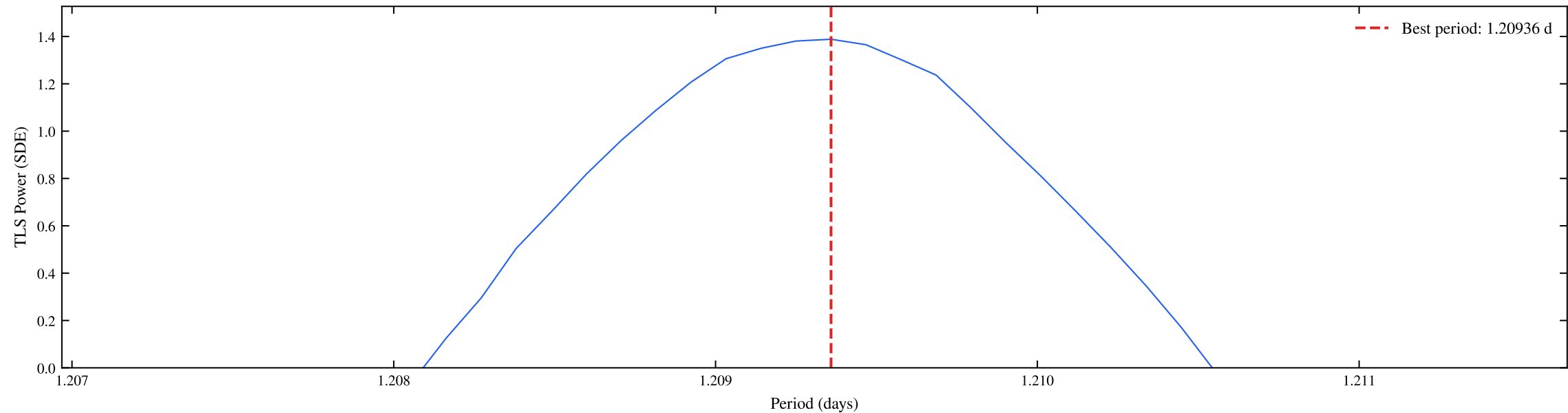
TIC 469782185 | Sectors: 67 | MCMC Trace Plots



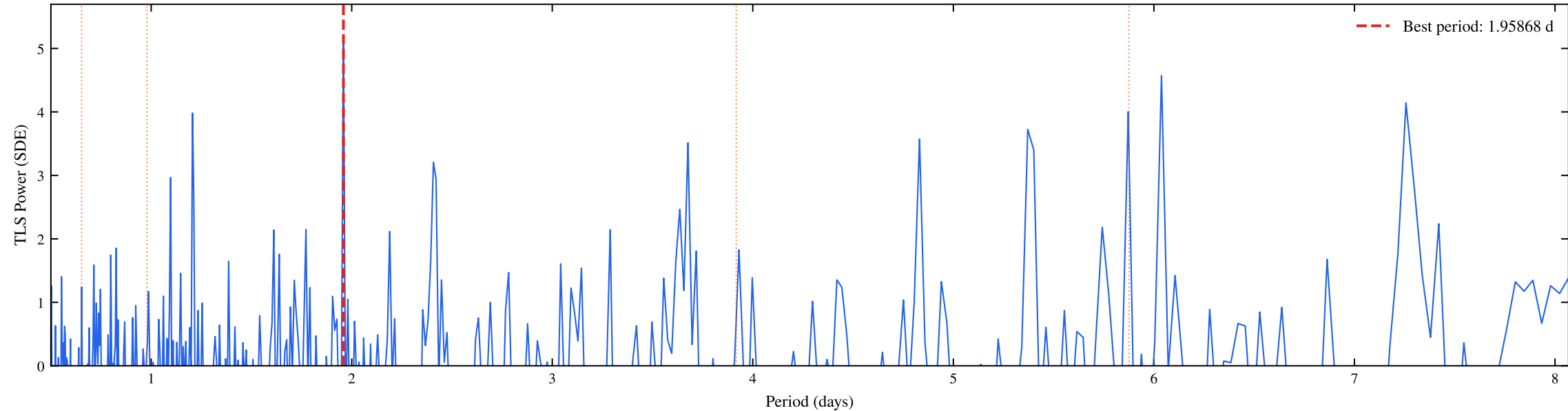
TIC 469782185 | Sectors: 67 | TLS Coarse Periodogram (SDE = 1.39)



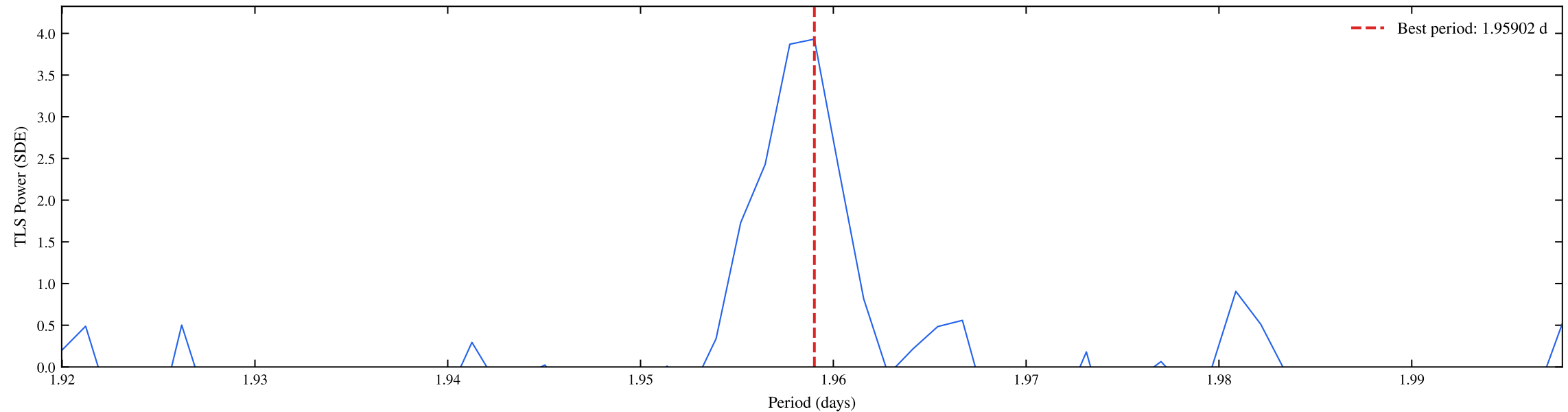
TIC 469782185 | Sectors: 67 | TLS Fine Periodogram (SDE = 1.39)



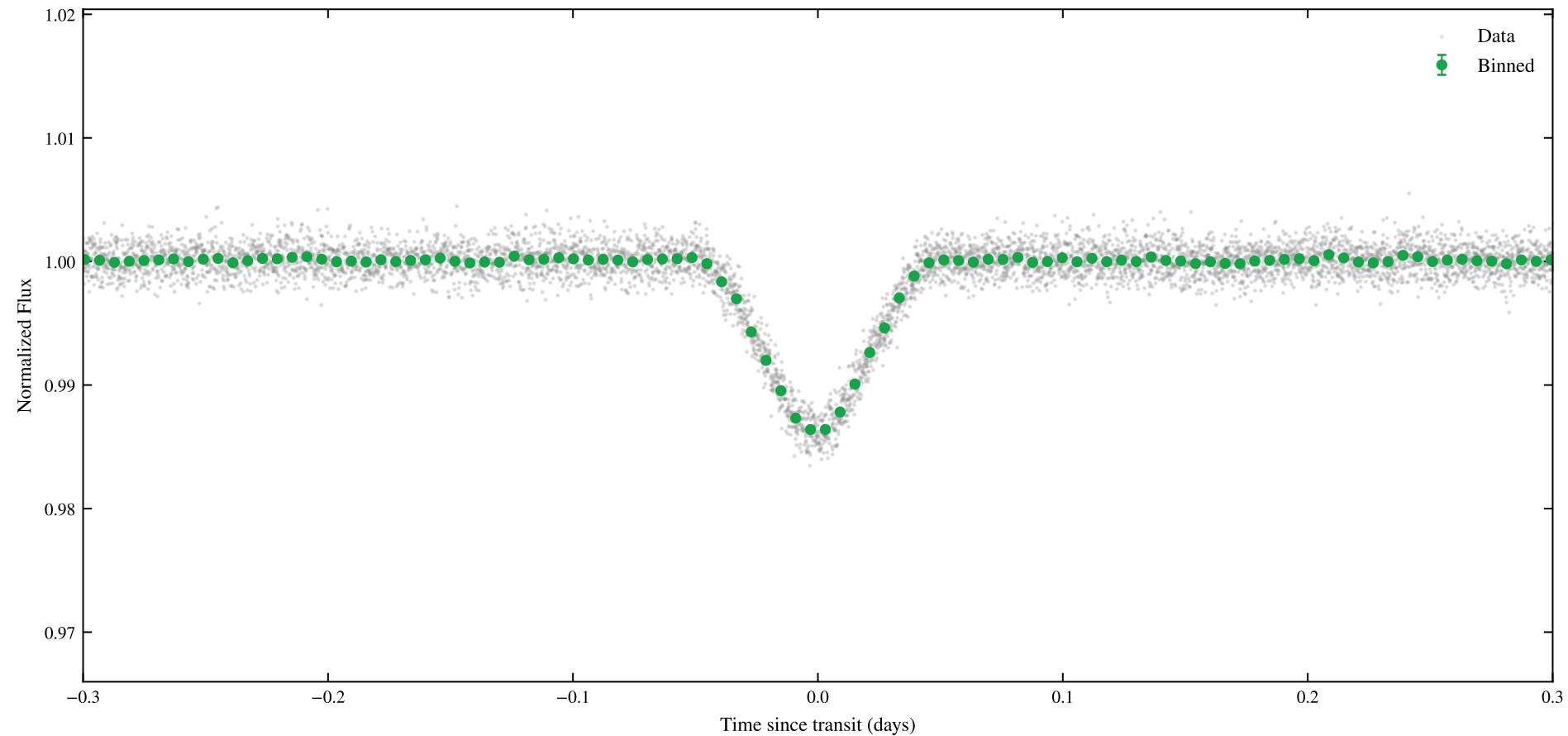
TIC 469782185 | Sectors: 67 | TLS Coarse Periodogram (SDE = 5.18)



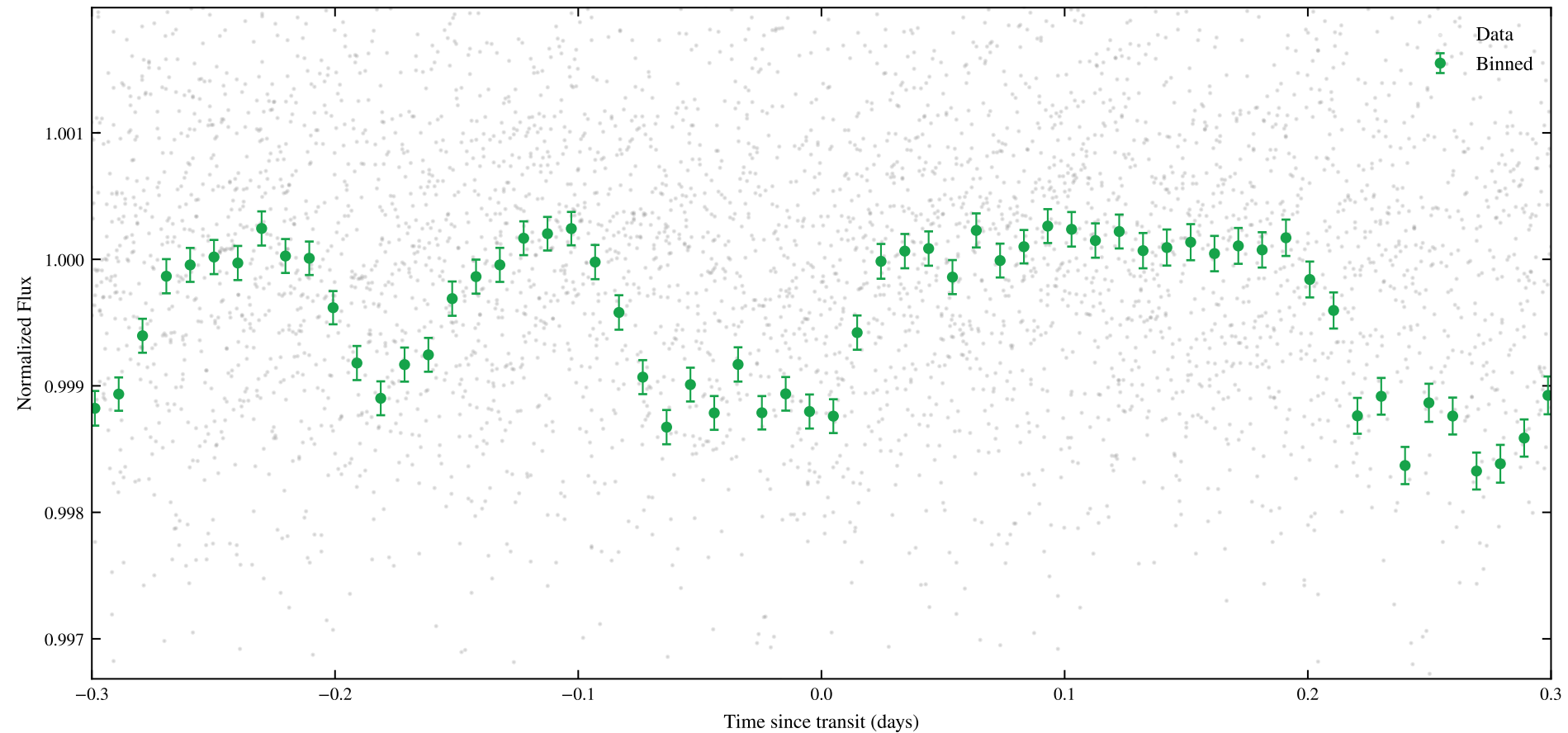
TIC 469782185 | Sectors: 67 | TLS Fine Periodogram (SDE = 3.93)



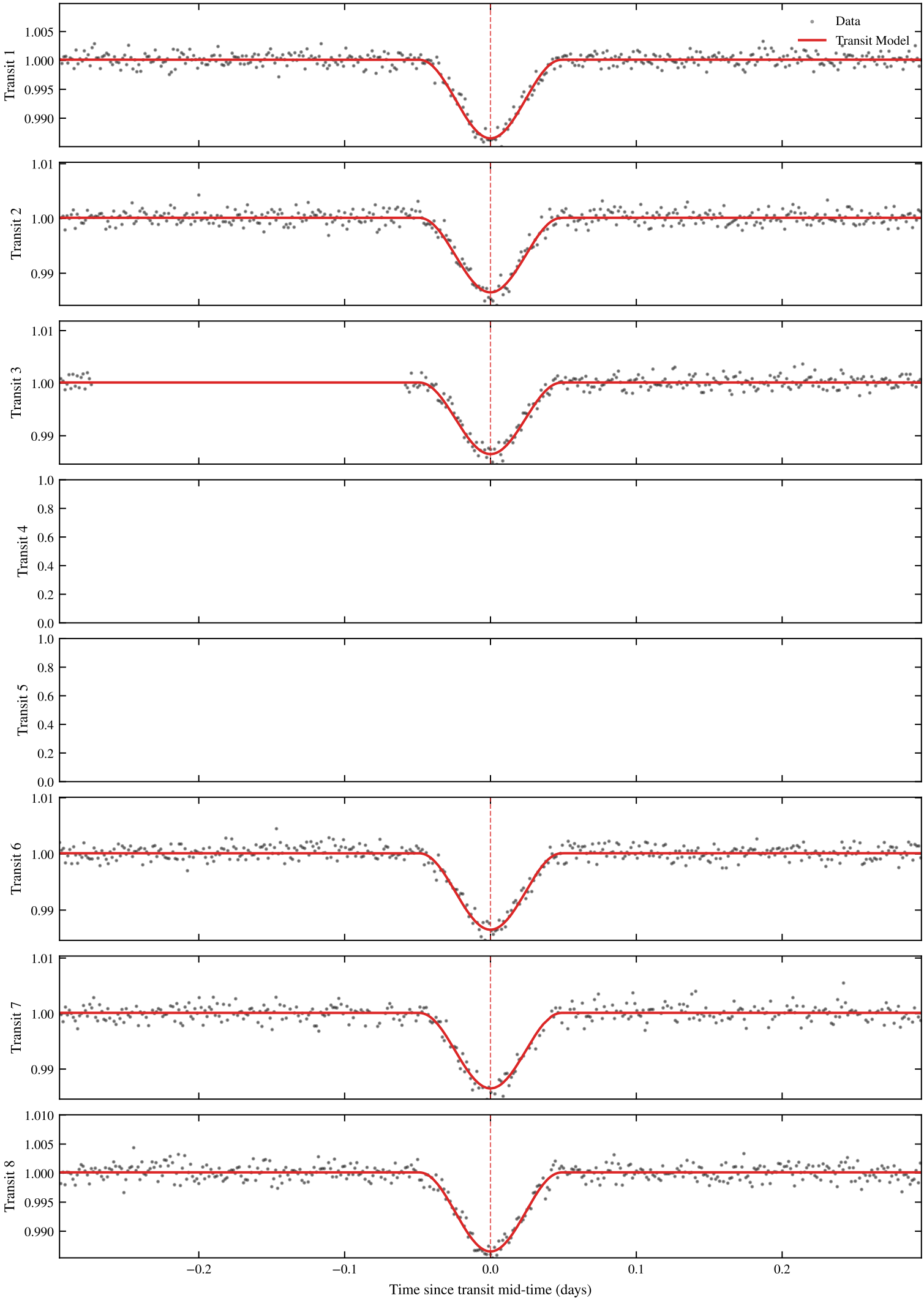
TIC 469782185 | Sectors: 67 | Phase-Folded Transit (P = 1.20936 d)



TIC 469782185 | Sectors: 67 | Phase-Folded Transit (P = 1.95902 d)



TIC 469782185 | Sectors: 67 | Stacked Individual Transits



TIC 469782185 | Sectors: 67 | Phased MCMC Fit - Planet 1

